

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T22 · Rearranging Formulas

With Answers

Questions with full worked solutions

■ 18 pages

FAMILY MAP

Rearranging Formulas

F01	Undo a single power or root in a formula	2 Q 6 marks
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F02	Rearrange a subject inside one fraction	3 Q 11 marks
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F03	Isolate a subject through inverse operations	3 Q 10 marks
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16 Make t the subject of the formula $c = \frac{t^2 + 3}{7 - 8t^2}$

(Total for Question 16 is 4 marks)

Worked Solution 27Make t the subject**UNIT 2****4 marks****Clear the denominator**

$$c(7 - 8t^2) = t^2 + 3 \Rightarrow 7c - 8ct^2 = t^2 + 3.$$

Collect and factor

$$7c - 3 = (1 + 8c)t^2.$$

Take square roots

$$t^2 = \frac{7c - 3}{1 + 8c} \Rightarrow t = \pm \sqrt{\frac{7c - 3}{1 + 8c}}.$$

Final answer

$$t = \pm \sqrt{\frac{7c - 3}{1 + 8c}}$$

Original question 01

4MA1-1H Q1 Q1(d)

ORIGINAL QUESTION**2 marks**

(d) Make t the subject of $c = t^3 - 8v$

(2)

Worked solution

Family: Undo a single power or root in a formula • Variant: Single squared/powered subject

METHOD / WORKING

MODULAR UNIT 2 • JANUARY 2022 DETAILED SOLUTIONS
Worked solution 01 MODULAR UNIT 2
Chapter: Rearranging Formulas 2 marks
Q1(d) – Chapter: Rearranging Formulas
Family: Undo a single power or root in a formula • Variant: Single squared/powered subject
Method
$c = t^3 - 8v$ gives $t^3 = c + 8v$.
Why: Add $8v$ to both sides.
Method
$t = \text{cube_root}(c + 8v)$.
Why: Take the real cube root of both sides.
Final answer
t: $t = \sqrt[3]{c + 8v}$

FINAL ANSWER

16 Make x the subject of $y = \sqrt{\frac{x+1}{x-4}}$

Worked solution

January 2020 | Question 16 | 4 marks

Family: Rearrange a subject inside one fraction | Variant: Subject in the denominator

METHOD / WORKING

1. *Square the equation* : $y^2 = (x + 1)/(x - 4)$.2. Multiply through by $x-4$ to remove the fraction.3. *Expand and collect x terms* : $x(y^2 - 1) = 4y^2 + 1$.4. *Divide by $y^2 - 1$* : $x = (4y^2 + 1)/(y^2 - 1)$.

FINAL ANSWER

$$x = (4y^2 + 1)/(y^2 - 1)$$

(b) Make c the subject of the formula $p = \sqrt{\frac{ac + 8}{3 + c}}$

(4)

Worked solution

January 2021 | Question 14 | 4 marks

Family: Rearrange a subject inside one fraction | Variant: Subject in the denominator

METHOD / WORKING

1. *Square* : $p^2 = (ac + 8)/(c + 3)$.

2. *Multiplythrough* : $p^2c + 3p^2 = ac + 8$.

3. *Collectterms* : $c(p^2 - a) = 8 - 3p^2$.

4. *Hence* $c = (8 - 3p^2)/(p^2 - a)$.

FINAL ANSWER

$$c = (8 - 3p^2)/(p^2 - a)$$

(b) Make t the subject of the formula $p = at - d$

(2)

Worked solution

June 2021 | Question 5 | 2 marks

Family: Isolate a subject through inverse operations • Variant: Linear add/multiply formula

Method / working

1. Add d to both sides of $p = at - d$: $p+d = at$.
2. Divide by a to make t the subject.

Final answer: $t = (p+d)/a$

(b) Make f the subject of $w = \frac{2f + 3}{8 - f}$

(3)

Worked solution

November 2021 | Question 17 | 3 marks

Family: Rearrange a subject inside one fraction | Variant: Subject in the denominator

METHOD / WORKING

1. (b)

2. $w = \frac{2f+3}{8-f}$

3. $w(8-f)=2f+3$

4. $8w-3=f(w+2)$

5. $f = \frac{8w-3}{w+2}$

FINAL ANSWER

$(6y+5)(y-1), f = \frac{8w-3}{w+2}, 4(x-1)^2 + 3$

15 Make x the subject of $y = \frac{5 - 2x}{x + 3}$

(Total for Question 15 is 4 marks)

Answer reference | May-Nov 2020 | Question 15

Family: Isolate a subject through inverse operations • Variant: Linear add/multiply formula

15	$xy + 3y = 5 - 2x$ oe			M1	multiplying both sides by $(x + 3)$ and expanding the brackets correctly
	e.g. $xy + 2x = 5 - 3y$			M1	ft dep on 2 terms on left and $(5 - 2x)$ on right, for collecting all x terms on one side and non- x terms on the other side
	eg $x(y + 2) = 5 - 3y$			M1	ft, dep on 2 terms in x , for factorising for x
		$x = \frac{5-3y}{2+y}$	4	A1	oe allow $\frac{5-3y}{2+y}$ as answer so long as previously seen $x = \frac{5-3y}{2+y}$
					Total 4 marks

Worked solution

May-Nov 2020 | Question 15 | 4 marks

Family: Isolate a subject through inverse operations • Variant: Linear add/multiply formula

Official final mark-scheme pages are embedded immediately after this question.

18 Make y the subject of $p = \sqrt{\frac{y + w}{3y - t}}$

(Total for Question 18 is 4 marks)

Worked solution

November 2025 | Question 18 | 4 marks

Family: Isolate a subject through inverse operations | Variant: Subject in a product with several factors

METHOD / WORKING

Worked solution
November 2025 Question 18 4 marks
Family: Isolate a subject through inverse operations Variant: Subject in a product with several factors
Solution
$p = \sqrt{(y+w)/(3y-t)}$
$p^2 = (y+w)/(3y-t)$
$p^2(3y-t) = y+w$
$3p^2y - p^2t = y+w$
$y(3p^2-1) = w+p^2t$
Answer: $y = (w+p^2t)/(3p^2-1)$
PROVISIONAL WORKED SOLUTION — NOVEMBER 2025 QP-ONLY EXCEPTION

FINAL ANSWER